Earth Curvature and Refraction in the GoSurvey Traverse Editor

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Abstract

GoSurvey's Traverse Editor reduced a leg's vertical angle to an elevation difference (ΔZ) by plain trigonometry alone. Over any leg of meaningful length this is not what a level line actually measures: the line of sight bends slightly toward the ground through atmospheric refraction, and the reference surface itself curves away beneath it. This note derives the standard combined correction, works it by hand for several leg lengths and refraction conditions, and compares each hand result against the value GoSurvey now computes. It also documents the automated test suite that keeps the implementation honest going forward, and records the resulting change in previously-computed traverse elevations.

1 What problem this fixes

A total station measures a vertical angle and a distance to a target, and trigonometry converts that pair into a horizontal distance and an elevation difference. Plain trigonometry implicitly assumes two things that are not true over any real distance: that the line of sight travels in a straight line through the atmosphere, and that "level" is a flat plane rather than a surface curving with the earth. Both effects are well documented in standard surveying references (Ghilani and Wolf, 2015; Bannister et al., 1998), and both act in the same direction on a normal (non-grazing) sight: curvature makes a level target appear to drop away below the horizontal, while atmospheric refraction bends the line of sight downward toward denser air near the ground, partially canceling the curvature effect. Left uncorrected, this is equivalent to always observing with a refraction coefficient of $k \approx 1$ no matter what the atmosphere is actually doing, an error that grows with the square of the distance and was reported against GoSurvey directly by a user running real traverse legs (GoSurvey Issue 826, 2026). This change adds the standard combined curvature and refraction correction to the Traverse Editor's ΔZ computation.

2 The governing formula

The combined curvature and refraction correction to an elevation difference observed over a horizontal distance D is

$$C \;=\; (1-k)\,\frac{D^2}{2R}, \tag{1}$$

where R is the mean radius of the earth and k is the refraction coefficient, the fraction of the curvature effect that atmospheric refraction cancels out (Ghilani and Wolf, 2015; Bannister et al., 1998). GoSurvey uses $R = 6{,}371{,}000$ m, the mean earth radius, and a default $k = 0.14$, the value standard references give for average atmospheric conditions (Ghilani and Wolf, 2015). C is strictly positive whenever $k < 1$ (the normal case): the corrected elevation difference is always *higher* than the plain-trigonometric value, for a level or an inclined sight in either direction, because curvature and refraction act on the vertical reduction independently of which way the sight is pointed.

For each traverse leg, GoSurvey already reduces the field observation (whichever of a direct horizontal distance or a slope distance plus vertical angle was entered) to a horizontal distance D and a plain-trig elevation difference ΔZ_{trig}. This change adds one term after that reduction, at the single point both input paths already share:

$$\Delta Z \;=\; \Delta Z_{\text{trig}} \;+\; (1-k)\,\frac{D^2}{2R}, \tag{2}$$

applied only when a vertical angle was actually resolved for the leg; a leg entered with no vertical observation keeps its legacy flat ($\Delta Z = 0$) behavior exactly as before. The option defaults on, since the uncorrected behavior is a systematic error rather than a deliberate choice, and can be switched off to reproduce the exact pre-change numbers.

Implementation. `src/traverse/TraverseCalc.cpp`, function `ComputeTraverse`, in the per-leg loop immediately after the horizontal distance is resolved and validated, before it is used for anything else (coordinate increments, closure perimeter, or the least-squares adjustment's inputs). `src/traverse/TraverseLeastSquares.cpp` performs no vertical reduction of its own, so this single point covers every elevation the Traverse Editor produces.

3 Worked example A: the textbook one-kilometer sight

A 1000.0000 m leg with a $+2^\circ$ elevation angle, at the default $k = 0.14$:

Quantity	Formula	Value
Horizontal distance D	measured	1000.0000 m
Plain-trig ΔZ_{trig}	$D\tan(2^\circ)$	34.9207 m
Curvature + refraction C	$(1-0.14)D^2/(2R)$	0.0674933 m
Corrected ΔZ	$\Delta Z_{\text{trig}} + C$	34.9882 m

This is the textbook anchor value: $C = 0.0675\,K^2$ for K in kilometers at $k = 0.14$ (Ghilani and Wolf, 2015), and $0.0675 \times 1^2 = 0.0675$ m agrees with the hand-computed 0.0674933 m to the

fourth decimal (the small residual is simply the published constant's own rounding). GoSurvey's test suite asserts this exact pair of values: `tests/TraverseTests.cpp`, *"Curvature+refraction: 1 km leg gains 0.0675 m at k=0.14"*, which isolates the correction by differencing a corrected run against the same leg with the option off, so the test is immune to any unrelated change in the plain-trig term.

4 Worked example B: refraction coefficient and leg length both matter

Because C depends only on D and k, not on the vertical angle itself, it is straightforward to check the formula's two free parameters independently against hand arithmetic.

Case	Formula	Hand value	GoSurvey
$D = 1000$ m, $k = 0$ (pure curvature, no refraction)	$D^2/(2R)$	0.0784804 m	0.078480 m
$D = 1000$ m, $k = 0.20$ (hazier air, more bending)	$0.80\,D^2/(2R)$	0.0627843 m	0.062784 m
$D = 300$ m, $k = 0.14$ (a short layout leg)	$0.86\,D^2/(2R)$	0.0060744 m	0.0060744 m

The $k = 0$ row is the full, uncancelled curvature drop, the largest the correction can be for a given distance under this formula; the $k = 0.20$ row shows the correction shrinking as k rises toward 1, exactly as Equation 1 predicts; and the 300 m row shows the term is already under a centimeter at ordinary traverse-leg lengths, which is why the uncorrected error was easy to miss on short legs and only became obvious on the longer legs the original bug report described (GoSurvey Issue 826, 2026). Each row is covered by its own case in `tests/TraverseTests.cpp` under the `[curvref][issue-826]` tag.

Sign check: downhill sights

Equation 2 adds C unconditionally, so a downhill ($-2°$) sight over the same 1000 m gets the identical $+0.0674933$ m correction as the uphill case in Section 3 — the corrected ΔZ is less negative than the plain-trig value, not more. This matches the physical picture: a level line of sight over a curved earth drops below the tangent plane regardless of which way the instrument is tilted, so the curvature term must raise the elevation difference in both directions. GoSurvey's test *"Curvature+refraction: downhill sights get a positive correction"* asserts both the magnitude and the direction of this effect.

The slope-distance path

A leg entered as a 1000 m slope distance at a zenith angle of 88° reduces to a horizontal distance of $D = 1000\sin(88°) = 999.3908$ m before the correction term is applied, giving

$C = 0.86 \times 999.3908^2/(2 \times 6{,}371{,}000) = 0.0674111$ m, which GoSurvey's *"Curvature+refraction: slope-distance/zenith path matches"* test confirms to the same precision. This matters because GoSurvey has two independent input paths for a leg (an entered horizontal distance, or a slope distance reduced through a separate function); the test demonstrates both paths reach the same correction code rather than only one of them being fixed.

5 The default (no vertical angle) path is unchanged

A leg entered with only a horizontal distance and no vertical observation — there being nothing to reduce — keeps $\Delta Z = 0.0000000$ m exactly, bit-for-bit identical to the pre-change behavior, confirmed by *"Curvature+refraction: legs with no vertical angle are untouched"*. This guards against a plausible implementation mistake: applying the curvature term to every leg's horizontal distance regardless of whether a vertical angle exists, which would silently invent elevation data for legs that never had any.

6 What changes for existing users

Because the option defaults on, elevations computed by the Traverse Editor for any leg that used a vertical angle will differ from earlier GoSurvey versions by exactly the C term above: roughly +6.75 cm per kilometer of leg length at the default $k = 0.14$, falling to about +6.1 mm at 300 m and +0.7 mm at 100 m. This is not a new bug; it corrects an existing one. Unchecking "Curv + Refr" in the Traverse Editor's options row restores the exact previous numbers, verified by the no-op test in Section 5 applied with the toggle left at its non-default (off) state.

7 Automated test summary

File	Test case	What it proves
`TraverseTests.cpp`	1 km leg gains 0.0675 m at k=0.14	the textbook anchor value, isolated from the plain-trig term
`TraverseTests.cpp`	k=0 gives pure curvature 0.0785 m/km	the uncancelled upper bound of the correction
`TraverseTests.cpp`	custom k=0.20 scales the correction	k is a live user input, not a baked-in constant
`TraverseTests.cpp`	300 m leg corrects 6.1 mm	the term stays correct, and small, at ordinary leg lengths
`TraverseTests.cpp`	downhill sights get a positive correction	the sign is direction-independent, as the physics requires
`TraverseTests.cpp`	slope-distance/zenith path matches	both leg-entry paths reach the one correction point
`TraverseTests.cpp`	legs with no vertical angle are untouched	the correction cannot invent elevation data from nothing

All seven cases, plus the eight pre-existing traverse test cases, pass under the `GoSurveyTests` executable (Catch2, MSVC release build), 15 test cases and 97 assertions with no regressions. The source is PR #839, branch `fix-826-curv-refr`, requirement entry pending in `spec/requirements.md`.

References

Bannister, A., Raymond, S., and Baker, R. (1998). *Surveying*, 7th ed. Harlow, UK: Pearson/Prentice Hall.

Ghilani, C. D., and Wolf, P. R. (2015). *Elementary Surveying: An Introduction to Geomatics*, 14th ed. Boston: Pearson.

GoSurvey Project (2026). *GitHub Issue #826: Traverse editor: add earth curvature + refraction correction to vertical angle reduction.* `https://github.com/chetjones003/GoSurvey/issues/826`